

# OffensiveCon24 - Piotr Bazydlo - Half Measures and Full Compromise

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## Content

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## Description

Half Measures and Full Compromise: Exploiting Microsoft Exchange PowerShell Remoting  
<https://www.offensivecon.org/speakers/2024/piotr-bazydlo.html>

## Transcript

*Retrieved with yt-dlp in a container, which is the only route YouTube still answers. Expect transcription errors in type and method names.*

[Music] So hi everyone, my name is P Bazeda. I'm a vulnerability researcher at the trend micro zero day initiative and today I'd like to show you my talk concerning uh exchange powershell remoting. Uh so in fact what you have to know is that Exchange PowerShell remoting is based on Windows PowerShell remoting. So you could say that it's like a

sandboxed PowerShell remoting with a couple of extensions. Um and a lot of stuff that I'm going to show you today is only applicable potentially to different applications that are using PowerShell remoting and in Exchange. The thing is that any domain user with a valid exchange networks or probably like any employee in your organization can invoke multiple commandlets of exchange powershell and typically this attack surface is only applicable to like internal attackers. So the attackers who are already in your network because you are connecting over HTTP but you require a valid kberos authentication. However, the history show that it was not always the case and for a long time actually you could access powershell remot exchange without a need to perform keros authentication. So the first example is the proxy shell chain by orange site. Uh so here he has found a very cool path confusion vulnerability which allowed you to abuse like exchange front end to proxy the request to them exchange back end and it didn't require any authentication. So you could in fact even reach the PowerShell without any authentication and then he has abused some of the logically issues if exchange comments and then one year one year later we had proxy shell chain which was exploited in the wild and those exploitation attempts have been found by GTSC who has submitted it to the ZDI. So we were analyzing this and it turned out that the path confusion hasn't been fixed. However, the attack surface the usage of this path confusion was closed for the pre-app attackers. Still you could like just add authorization header to your HTTP request use basic authentication NLM authentication whatever and you still able to exploit that. So you could still like reach the PowerShell back end over the internet without the keros and the second vulnerability in the chain was a wonderful derialization vulnerability in exchange power remoting. So before that I was not even aware of the attack surface but when I looked at it I thought that it's like really great attack surface and I decided to investigate. So once again any domain user in exchange with a mailbox can invoke multiple symbols. One of them is a get mailbox. So you provide some argument like identity here and you provide an input here. It's a simple input because it's a string. It's an email address. But if you would look up the documentation for any commandlet for exchange parro remoting, it turns out that you can provide more complex type like FQDN account partition ID parameter and couple of more of them. So it turns out that there has to be some underlying the serialization mechanism that is handing those more complex objects and there are some tips in Microsoft documentation for PowerShell remoting protocol. Basically it divides target types for the derialization in PowerShell remoting into two main categories. The first one are the primitive types like string, integer, byte array, a couple of more but also they say that you can deliver complex objects and here you have an example because everything that goes for PowerShell remoting is in form of XML. So we've got a serialized system drawing point object and some properties. However, the documentation doesn't say how the d serialization itself is been performed. So I had to go through the code for PowerShell remoting which was written in C# and I figured out something like this. So it's a very simplified scheme for PowerShell remoting this realization. However, it's like the main attack surface that I've been abusing and it is something that you can abuse in anything that's related to PowerShell remoting. So imagine that you are delivering some complex object in form of XML and you are specifying a target type. So we are saying that okay I want after the d serialization I want to have an object of this type and this type has to be

allowed you cannot des serialize anything and powershare remoting stores the right list of objects in ps1xml files so powershare remoting has its own default list of types that are allowed also exchange extends it so it adds the exchange related objects to the list with it and exchange specifies its own files like exchange types ps1xml So when the title is allowed, PowerShell remoting is retrieving some properties for the D serialization and at this point it tries to figure out the conversion mechanism to be used. So it takes the from type which you freely control. So you say okay I'm d serializing from a string. So the from type will be a string and the target type will be the your target type for the derialization. Let's say it's FQDN. So it calls the figure conversion method and it selects the internal derialization mechanism that it will use for the derialization and there are a lot of them. I have selected like three that are the most powerful ones. The first one is convert via parse method. So it allows you to call a public static parse string method of our target type. You've got convert via constructor which is probably the most dangerous one, the most powerful one because it allows you to call a single argument constructor of the type that you are the serializing as long as you can control the input argument to the constructor. So it is our from type like string and we've got a no argument constructor conversion which simply initializes the object with no argument constructor and uses setters to set the values um for the members. The very important thing is that the figure conversion method is a bunch of if statements. So it goes over the conversions in an order and it selects the first one that is applicable, right? And then it performs this realization. Also what is very important is the fact that PowerShell remoting allows you to perform some extensions. So it allows you to define custom converters and exchange has its own custom converter is called serialization type converter. So if you would look at some sample type definition in PS1 XML file, here you've got an sample for exchange. It specifies that the target type is IPvx address and you can see a definition for a custom converter for the exchange and right now we have to say what this exchange serialization type converter does. So basically it's based on a binary formatter the serialization of net and in general it's dangerous to use that. So you have to implement a valid white list and block list of classes in order to be protected and not get pawned very easily. Of course, exchange does that. It has a white list, black list and so on. So you cannot like easily pawn this at this moment. And now we go straight to the pract. So as I have already mentioned, it is a very complex vulnerability. I have described it on the ZDI blog post like one year ago. uh I've provided for full payload and the call flow. So if you are interested into the details you can have a look but here I'm going to show you the fundamentals which may allow you to like fully understand that. So let's imagine that you are delivering some complex object and the target type is like service controller and within your object you are delivering two main things. The first one is a property called target type for this serialization. So it contains a binary formatter serializ serialized sorry unity serialization holder object and that's the root cause in fact for the entire proxy in a nutshell that this unity serialization holder is whit listed in exchange custom converter. So PowerShell remoting will firstly try to derialize this property. We can force it to use this exchange custom converter and it des serializes it. And it turns out that if you des serialize Unity serialization holder, you're retrieving an object of type type. So you're just retrieving a type and you fully control that. And here we are retrieving a type example reader. If you are dealing with net, it may ring a bell. So it

the here the neat trick about the powershell remote was that it allows you to override the target type for the entire powershell remote in this realization with the type that you have retrieved with this exchange custom converter. So basically what we are reaching here we are able to have internal powershell remote in this realization reach with any type that we want. So the white list does not longer exist for us and the next steps are very easy because we are delivering a from type which is string it is just exam rce gadgets for net so power share remote will say okay so I'm deserializing from a string to a reader right and I have this parse based conversion which will just call a reader parse method and it's like an rce primitive in net right so it is an easy remote code execution through the the serialization of exam. So basically just to sum up the this great vulnerability uh that was exploited in the wild you can use it to deserialize anything you want with those internal powershell remote serialization routines. So after analyzing this, I've decided to have a deeper look into the entire attack surface for PowerShell remote in exchange this realization and it led to like 14 CVAs, a couple of RC chains, file rights, file reads, NLM relays because at that point NLM relays were very dangerous in exchange. They allowed you to escalate privileges. So like Microsoft was treating them very seriously. And a couple of more researchers also have found some like exploitation gadgets for that. So majority of these stuff that I have found was based on the idea that you can like override the target type and do some fancy stuff but not all of them. And I'm going to show you them because they are they may be important if you ever see like PowerShell remoting used somewhere else. And here we've got a very fun like concept for PowerShell remoting. So exchange has right listed a file info class for powershell remoting and file this serialization of file info in net list to antl relay right on the other hand you can see that there is this exchange custom converter was specified for it and when you look at this converter it has an internal white list and it turns out that file info is not whitelisted inside so at this point you may say okay it's all right because I will not be finally able to deserialize file info. But what I have already mentioned is that conversion order matters in PowerShell remoting. So even if you allow some class to be deserialized with your custom converter, you have to be aware of the fact that you the custom conversions are like last resort. So PowerShell remote will firstly try to deserialize them with its own conversions. So here the payload was very simple. So I'm just like delivering target type which is file info. I'm providing a string and powershell remoting will say okay I can just use a single argument constructor of file info to the serialize that and it leads to antl relay. So this is something that you have to be aware of when looking at partial remote industrialization. And now let's get back to practice in a shell. So we submitted to Microsoft. I was waiting for the patches and I was thinking about the potential patch. And for me the clear thing is that was that they are probably going to introduce some another internal white list that provides another type control for this like unity serialization holder. So my idea for the research was to look for like custom deserialization gadgets in exchange code base. I have found like a couple of RCA gadgets, file read gadgets and so on. And I thought that maybe some of those gadgets will be included in the white list because they will turn out to be like important for Exchange PowerShell. And I was right somehow because majority of those gadgets were treated as duplicates by Microsoft but two of them were treated as unique issues. one NLM relay and one remote code execution. So you could say that I have bypassed like the patch for the proxy nutshell

before it even appeared because I have reported to Microsoft they have released the patch for the proxy nutshell but my rce was still hitting because one of the classes that I have used uh for the exploitation was um whit listed. So before moving to the gadget itself let's have a look at the patch. So this unity serialization holder that allows you to d serialize a type is still allowed. But as I have already mentioned they implemented another like type control internal type control. So you cannot d serialize and you cannot override the target type at this point with anything you want. But the bad luck was that one of the gadgets that I have found was allowed and it was not a typical gadget. It was a very powerful gadget that I have called something like a bridge gadget and it was like a fundament for my future exploitation attempts. So a quick look at the gadget. Uh it was called multivalued property. It's one of exchange classes and it's a generic class. It has a single argument constructor that we can use with those internal powershell remoting conversions. And the thing is that if we d serialize it with this exchange custom converter, we fully um control the internal type because it's generic class. So you can specify something like multival property of string of FQDN or any other type and this generic type can be in fact treated by you at this point as an internal target type because this constructor of multival property leads to another internal derialization mechanism. This time it's implemented in Exchange and that in PowerShell remoting, but it's very similar because it implements two powerful conversions exactly the same like the ones that you have in PowerShell remoting. The one that allows you to call like a public static parse string method and a single argument constructor and we are calling it on this internal target type that we fully control again. So if you would look at the scheme for this, we've got u the difference between this one and the pract from the attacker perspective is this serialized target type for this serialization. Previously it stored exam reader type and now it stores multival property of exam reader. So powershell derializes it uses this exchange custom converter. We are retrieving multival property of exam reader and at this point after the patch exchange exchange ch checks this type right and it says okay multival property is right listed so it's good but it never verifies this internal types of the exam reader so in fact we are overwriting the target type with the multival property of exam reader and then in the next step we are d serializing from string to multival property of reader action change PowerShell remoting says okay I can use a single argument constructor conversion and it leads to this internal this serialization mechanism when we have from type string still but the internal target type is exam reader and again we are calling the exam reader parse method which is like rce primitive and basically uh that's all so in fact we have reached something similar that we have reached with pract in a nutshell but we have just abused that whit listed classes of exchange which is like frequently used in powershell remoting and it implements another derialization mechanism which had no type control. So basically the same story but in a different way. Okay. So after that I was waiting for the patches and I was expecting patch to implement another type control this type time for this internal derialization mechanism. What I didn't expect it that it is going to be ex based on the blacklist. So they didn't use a white list for this internal derialization mechanism but blacklist. So you can still call a single argument constructor for instance for anything that isn't blacklisted. and exchange blacklist is very extensive. It contains a lot of classes. But exchange codebase is also very huge and net is huge. So there is always a

space to look for something more. And basically I bypassed this patch immediately because as you may remember I have found a lot of gadgets before the proxy nut shell was patched. I reported them to Microsoft and they were treated as duplicates majority of them but in fact they have never done anything with them. So they didn't include them in the blacklist. So I've just took one of the gadgets that I have found like 6 months before. I have chained them with this internal multivalued property derialization mechanism and it was just an easy RC for me. So the gadget was a command class. We are chaining it with multival property and what you deliver to the constructor of command is just getting executed. So you're just starting a new process, right? So here I'm providing cmd exit to the constructor and it's getting executed right and that was it at this point. Okay. So after that I'm still waiting for the patches right and I was surprised but the patch extended the blacklist with all the gadgets that I have already reported. So basically that's why this talk is called half measures and full compromise because like extending the blacklist with the gadgets that were reported to you may not end pretty well and in fact NLM relay and denial of service gadgets were so easy to find at this point you have like dozens of net classes that you can call a single argument constructor and allow us to perform some operations on the file. So I wasn't even focusing on them at all. So I just decided to go straight for remote code execution stuff. And very important thing at this point is that the path confusion that allowed you to like attack exploit this attack surface from the internet has been closed like even before dispatch. So right now you need to be like internal attacker. So because you need a valid kerberos authentication. Uh so you need to be inside the domain but still any employee can get system on the exchange. uh um any internal attacker that got access to your domain can get an RC on exchange so it's not very good right and at this point looking for gadget was not so easy but there were still some to be found and the first one was exchange class called approved application collection and it it single argument constructor accepts a path to the cabinet file so as I have already mentioned you have to be inside of the domain in order to even exploit that at this point So you can just create a public sambar share pull the cabinet file over there deliver path to the to this constructor and exchange will try to extract it to some temporary directory. The issue was that it was using extract 32 which is windows utility and this windows utility was just vulnerable to path traversal. It's like a zip slip but for cabinet files. So the idea was very simple. you are putting a path traversal sequence into the cabinet file, you're uploading a web shell. The only issue was that if you are putting like dot dot back slash sequences, it was detected by the defender. But it turned out that if you like replace the backslash with a forward slash, defender is not detecting that anymore. So it was like a bypass for the defender and that's it. You are holding a web shell and you're good to go. And one important thing is like we have reported this extract path traversal in Windows to Microsoft and they say that it's not no fix right. So it's like a caller's responsibility to verify that you are not extracted untrusted cabinet files. It's not fixed. It still hits. So if you ever see an application using extract 32 for cabinet file handling, you know that you cannot use that. Uh okay. So at this point I knew that they are going to probably extend blacklist with this gadget. So I decided to look for one more and it led me to one of the most interesting chains that I have ever ever found. like it was very hard to exploit it at this moment for me. And I have found an exchange class called SH type resolver. Again, we are abusing single

argument constructor. So you can see that it was really one of the most powerful conversions and this gadget in fact accepts a path a string and it tries to load a DLL called fuse packos DL from the path that you provide. So you have to know that since net 4 remote DL loading is prohibited by default, right? So you can't it's not an rce gadget at this point. You cannot just create some basher in the domain and load the DL remotely. It won't work. It's a local privilege escalation gadget at this point. It's pretty good, but I still wanted to make it RCA because it looked cool. And I thought that I may need to look for some like firewrite primitive to drop a DL to the file system and then load it. And it turned out to be very hard task. But it led me to another like exchange internal class which was called dump data reader. And what it does it you again provide a path to the cabinet file. It verify if it exists. And after that it creates a temporary directory which you could say is random because it's based on GUID and you cannot predict the GUID. Then it tries to extract it with another Windows utility which is Xband and Xband is not vulnerable to patch traversal. And also you can see this F switch which says that we can only extract files that end with DMP and we have to extract fuse Paxus DL right and after that it's performing some operations and it's removing the files. So at this point I thought that it's completely useless because you've got too many issues and it completely doesn't match the requirements that you need to do like you need to upload the DL the file gets removed and all that stuff. After a while though I realized that this expand call is vulnerable to argument injection like it the path that you're providing is not verified in any way. And so I started playing out with expand and the first thing that I have discovered is that if you inject I or R switch it completely ignores this filter for the extension. So here you can see that I have appended I argument and I have um extracted file with a different extra extension than dmp. So that was the first thing. Second thing was that if you provide this error probably from recursive switch it unpacks entire directory structures. So you can have a cabinet file which has a directory and DLL inside of this directory. It extracts it but this file valation routine that is implemented in this exchange gaz jet is called with recursive set to false. So it will never remove our DLL. So DL still stays on the file system. And another thing is that you can provide multiple cabinet files to a single expand execution and it will handle all of them. So it led to a payload like this. We've got three cabinet files and an R argument injected. The only small issue is that such a file needs to exist because Exchange is verifying if the file exists or not. So the sambar structure is a little bit tricky but still easy to do. But at this point you may notice that we are still missing one thing. So we don't know the GUID because our local DL loading gadget requires you to provide a full path. path is based on the GUID and we don't know the GUID. We have to leak it somehow and that's why we have three cabinet files here. One of them is corrupted. And by corrupted I mean it contains a file with an invalid file name because I've discovered that when Expand tries to extract it, it has an exception and exception is been written to one of the Windows log files and this exception stores the entire path when it was trying to extract the cabinet file. So in such a way we can leak this GUID. So at this point I checked the exchange and net codebase again found an xxe gadget because again we are having a black list and not white list so it's easy to look for stuff and in such a way I was able to look the GU ID. So I would like to show you a demo of this gadget because of this chain because it's like my favorite chain. So on the right side I've got an exchange machine. On the left side we have

attackers machines and I'm starting the exploit and in the first stage you can see we have a file right because we are creating temporary directory and DLS are inside. So in the next stage I'm using xxe to leak the GUID from Windows log files. So the GUID has appeared in the bottom of the screen. I'm just copy pasting it to the exploit and we are loading the DLL and we are popping a calc. So that's it. So again waiting for the patches and this time the patch was very good because the entire weight listing mechanism for those exchange custom conversions was at this point only based on allowed lists and the internal types that we are providing for the generics are verified too at this point. So we are no longer able to abuse this internal derialization mechanism multivalued property in order to do the fancy stuff and the serialize anything that isn't blacklisted. It doesn't work anymore. So just to finish this research I decided to focus on allowed classes again and it led me to the thing that I have missed in the beginning. So this convert via no argument constructor conversion for the powershell remoting turned out to be more dangerous and more fancy than I had thought in the beginning. So what it does um imagine that you are providing some whitelisted type to the powershell remoting it serializes it with this no argument constructor conversion. So it just initializes the object with no argument constructor of course but then it tries to go over the properties public properties of our whitelisted type and it tries to d serialize them again with those internal powershell remoting the serialization routines but they don't have to be whitelisted right so you've got like whitelisted class and then it has not whitelisted members and powershell remoting will d serialize them right so it's a massive extension to the entire attack surface which I wish I knew before but yeah that's how it works. uh so when it performs the d serialization of members it just sets them with a setter right so again it's a massive extension to the attack surface and at this point I said okay I will go through the whitelisted classes for exchange and verify if I could abuse that and it turned out that I had found three more vulnerabilities here one um xx that led to fire read and to NLM relays and it looks like this you've got an allowed federation trust in exchange but it has a notallowed member it's called orc certificate and the type is x5009 certificate 2. So what you do in the payload you define a set of members one is called orc certificate and you say that we are d serializing from a string here we just provide the UNC path that we want to abuse and it will call the convert via constructor method of this x5009 certificate tool which is not whitelisted but you can still derialize it because it's just a member of whitelisted class. So another extension of the attack surface for powershell remoting that you have to be aware of while targeting it. Another example is xxe. Um you've got an allowed transport config container class and here we are not abusing the d serialization possibilities itself because it's just a simple setter call uh with a string but the setter itself leads to xxe right exchange is running on the old versions of net. This is also the setter call possibilities are something that you have to be aware of. Okay. So that would be all for the vulnerabilities. I only want would like to summarize the current state for this. So you have this better binder which is based on the white list and there was also one very strong um restriction of attack surface that was applied. So right now if you are targeting exchange custom serialization converter you need to sign your payloads. You've got you need to have a certificate and the payloads need to be signed otherwise they will be dropped. I haven't verified the signature mechanism by myself so we can have a look but it's a huge restriction to the attack



surface at this point. Still you should be able to des serialize anything that uh is whit listed in this PS1 XML files but only with those PowerShell PowerShell based conversion. So the one from the exchange will not be applicable without a certificate at least it shouldn't. Uh, okay. I will skip gadget searching tips because uh I've just run out of the time. But if you have any questions, I can answer them uh offline and also we will publish this stuff uh soon. So yeah, thank you very much for your attention. Hope you enjoyed.

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